BATU ÖNLÜKUŞ

batu.onlukus.20@gmail.com | +90 532 510 5884 | linkedin.com/in/batuonlukus | github.com/batuonlukus | batuonlukus.github.io

Third-year AI and Data Engineering student focused on machine learning and data science for finance: generative models for financial markets, credit-risk modelling, inflation measurement from large-scale price data, and LLM systems.

EDUCATION

Özyeğin University | B.Sc. in Artificial Intelligence and Data Engineering — Istanbul

Third year · Expected graduation 2028 · GPA 3.32/4.00 · 50% YKS Scholarship

Honor List in all four semesters, including High Honor once

Coursework: Intro to Machine Learning, AI in Finance, Database Systems, Data Structures & Algorithms, Probability & Statistics, Linear Algebra

BJK Kabataş Science and Technology High School — 2019 - 2024

RESEARCH EXPERIENCE

Undergraduate Researcher | Generative Models for Financial Markets — Sep 2026 - Present

Özyeğin University · Supervisor: Emre Sefer

- Diffusion-based generative models for limit order book data, in a three-person student team
- Reproduced reference experiments, ran follow-up experiments with pass criteria fixed in advance and set up the team's research repository; now working on multi-asset order book modelling toward L3 data

Lead Contributor | Web-Based Inflation Monitoring Project — Mar 2026 - Present

Özyeğin University · Supervisor: Olcay Taner Yıldız

- Ongoing study: daily prices from 100+ Turkish online retailers in 13 categories feed a CPI-weighted price index
- Lead contributor in a 23-person team: plan data collection and direct members on which sources to build
- Top contributor to the team codebase (900+ of ~4,200 commits), with scrapers across several retail categories

EXPERIENCE

AI Fellow (Part-time) | Özyeğin University — Oct 2026 - Present

- Selected for the university's AI Fellows program; designing and building AI-driven solutions for the university's institutional and administrative processes

Cloudpeer Global Teknoloji A.Ş. — Teknopark İstanbul

R&D Software Engineer (Part-time) — Aug 2026 - Sep 2026

- Designed and built Jetty, a local-LLM-based intent routing system for Jetimza, the company's digital signature product; owned the full design and architecture and handed it over to the team
- Answers only from approved help content, abstains under uncertainty and rejects out-of-scope or malicious requests; re-evaluated after every change on a hand-written synthetic test set

Data Science Intern — Jul 2026 - Aug 2026

- Built a machine learning based PD model on IDES, the credit-risk and guarantee platform Cloudpeer supports for İGE, with WOE binning and logistic regression so every score carries its reasons; presented to the CEO and CTO
- Built a monitoring layer for incoming applications (PSI drift, variance and covariance shifts, unrecognised profiles), tested on synthetic failure scenarios, with a local LLM writing the reports

Developer & Chain Magazine Coordinator | ÖzÜ Blockchain Club — Jun 2024 - Jan 2026

- Contributed to smart-contract development, internal tools and technical workshops; coordinated Chain Magazine, a weekly research and knowledge-sharing effort on blockchain protocols and tokenomics

PUBLICATIONS

U. K. Güneş, O. K. Demirtaş, ..., **B. Önlükuş**, ..., O. T. Yıldız. "A CPI-Weighted Web-Based Price Index for Monitoring Inflation and Cost of Living in Türkiye." ASYU 2026. Co-author and one of the lead presenters at the conference (Sep 2026).

HACKATHONS

- **Granit** | Kapsül Teknoloji Blockchain Hackathon, 3rd Place (2024): building-inspection records as NFT-based identities with IoT integration on Polygon (Solidity, Hardhat, React)
- **IDO Platform** | Units Network Hackathon (2024): tokenization platform for startup funding where contribution share maps to voting power (React, Ethers.js)

TECHNICAL SKILLS

Programming: Python, SQL (SQL Server), Java, Solidity
Machine Learning: scikit-learn, LightGBM, XGBoost, logistic regression, WOE binning, SHAP, AUC-ROC, PSI, OOD detection
LLM Systems: local LLMs, intent routing, retrieval and ranking, evaluation set design, safety filtering
Data & Tools: pandas, NumPy, Matplotlib, web scraping, scheduled pipelines, Git, Jupyter
Blockchain: Solidity, Hardhat, Ethers.js, Remix
Languages: English (fluent), German (basic)